

Membrane Air Separation

Final Lab Report

Unit Operations Lab 1



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Team 3

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Executive Summary:

Air separation is very crucial for industrial applications. It plays a significant role in the production of steel, petrochemicals or use it for clean energy purposes [1]. Air separation techniques like membrane technology occurs at non-equilibria. Therefore, it is important to study the separation system and observe how change in parameters (pressure and flowrate) vary the kinetics of the air flow into the system.

In this experiment, two Prism™ separator modules [2] are used to separate oxygen from the dry air components such as nitrogen and other inert gases. Each membrane consists of thousands of micro-fibers that act as a filter for the molecules [1]. As the pressure and the inlet flow is increased, the air with high pressure enters the separator and components are selectively separated. This allows oxygen to permeate across the walls of the membrane. However, nitrogen exits from the end of the shell. The oxygen sensors at the exit can read the concentration of the oxygen exiting out of the stream. The Prism™ separator modules are placed in parallel and series configurations to observe the flow of air through the system. The shell and tube side flow rates were also measured and recorded.

To study the flow of oxygen and nitrogen through the fiber wall, the overall mass transport coefficients were derived for both, N₂ and O₂, using the flow rates, shell and tube pressures of each gas respectively. While approximating tube side oxygen partial pressure, inlet and outlet pressure is averaged [2]. The separation coefficients, experimental separation factors and recovery of oxygen and nitrogen were also calculated and reported as a function of operating pressure [2].

It was observed that transport coefficients and oxygen selectivity was higher in series configuration as compared to the parallel configuration. This indicates the relation that higher the time of contact better will be the separation of oxygen across the membrane. Also, oxygen recovery was observed to increase at increased pressures however, decreased with increase in flow rates. The latter observation further signifies that diffusion flux of oxygen across the membrane relates to the time of contact.

Introduction:

Polymer based membranes are a very useful and economical tool in the chemical engineering industry. These membranes may be used for many mixture fluid separations without modifying the chemical structure or equilibria of such fluids. It is important to note only material transfer will occur as a selectivity of the solution based on diffusion and solubility. A popular separation that is used in industry is oxygen to nitrogen from air. Nitrogen may act, “as an inert gas of low reactivity, it is used as a protective gas in many sectors of industry; in the food industry or in the aerospace industry to inert the fuel tanks of planes and create a flame-resistant atmosphere” [4]. It is known that air is rich in nitrogen about 78% and 21% oxygen with a trend of high selectivity to low permeability and vice versa. In industry if the trend can be manipulated with the correct material then a pure and efficient separation can occur.

Focus on selectivity will be analyzed among multiple Poly-Sulfone miniature membranes that form as one large membrane fiber. With a system of Air passing through a polymer membrane a percentage of Oxygen composition may be read from a meter. The importance of the percentage meter at both the tube and shell side is beneficial due to simultaneously obtaining the molar fraction of oxygen and nitrogen in air. With the molar fractions known operating pressures and flow rates calculations may be made to find partial pressures, permeation rates, transfer coefficients and selectivity. A key advantage to using a membrane is low energy separation (no phase change) and from the calculations one may determine if incomplete separation occurred; in which the system would need a higher selectivity membrane [3]. Manipulation of operating pressures among parallel and series modules will be analyzed to better understand the relation of separation among membrane configurations.

Theory:

Gas permeation is a process involving membrane separation that uses a nonporous semi-permeable membrane. Membranes may be in the form of hollow fibers, “In general, hollow fibers are preferred as they achieve a higher effective membrane area within a given module volume”. [5] In this experiment, the gas feed used was separated into a permeate stream and a non-permeate stream. A solution-diffusion mechanism in which the streams were separated as “tube” and “shell” and therefore separate calculations could be made. In which the overall objective is to calculate mass transfer coefficients of oxygen and nitrogen. The gas molecules go through the membrane air separation unit consisting of 2 separate modules in parallel and series configuration separately. In gas, each component present has a unique characteristic of permeation rate which is dependent on diffusion and solubility. the two main concepts used in this experiment to calculate the coefficients are: Henry’s Law and Fick’s Law. For this lab, the two main components of interest were oxygen and nitrogen. The mass transfer coefficients of oxygen and nitrogen were calculated by:

$$O_2 \text{ permeation rate} = KO_2 \times (\text{tube side } O_2 \text{ pressure} - \text{shell side } O_2 \text{ pressure})$$

$$N_2 \text{ permeation rate} = KN_2 \times (\text{tube side } N_2 \text{ pressure} - \text{shell side } N_2 \text{ pressure})$$

The permeation rate was obtained from the shell side flow rate of O_2 and O_2 %level.

$$O_2 \text{ permeation rate} = (\text{Shell side flow rate of } O_2) \times (O_2 \text{ \%level})$$

$$N_2 \text{ permeation rate} = (\text{Shell side flow rate of } N_2) \times (N_2 \text{ \%level})$$

The main concept to keep in mind was the selectivity of the binary system. It was calculated using

$$\alpha = \frac{Q_{O_2}}{Q_{N_2}}$$

Where Q_{O_2} is the permeability of oxygen and Q_{N_2} is the permeability of nitrogen. To obtain selectivity, the permeabilities of each component were calculated using the Fick's law and correlations:

$$J_{O_2} = Q_{O_2} (P_{TO_2} - P_{SO_2})$$

Where, J_{O_2} is the flux of O_2 and P_{TO_2} is the tube side pressure and P_{SO_2} is the shell side pressure. The oxygen sensors measured the oxygen contain in mole fraction, so it was important to determine the separation of the two components in two different configurations i.e. parallel and series. The separation factor was evaluated by:

$$\alpha'_{AB} = \frac{\frac{\chi_{Ap}}{\chi_{Bp}}}{\frac{\chi_{Ar}}{\chi_{Br}}} \quad \text{Separation factor based on retention configuration}$$

$$\alpha''_{AB} = \frac{\frac{\chi_{Ap}}{\chi_{Bp}}}{\frac{\chi_{Af}}{\chi_{Bf}}} \quad \text{Separation factor based on feed composition}$$

Results:

Figure 1. clearly indicates that as the flow rate increases the level of oxygen also increases for both parallel and series configuration for the shell side. The difference in oxygen levels is not dramatic however, for the shell side, level of oxygen increases with operating pressure. This relation was found to be inversely related for the tube side where oxygen level is observed to decrease with the increase in operating pressure for series and parallel configuration. Therefore, it was concluded that higher amounts of oxygen permeate across the membrane from the shell side with the increase in operating pressure.

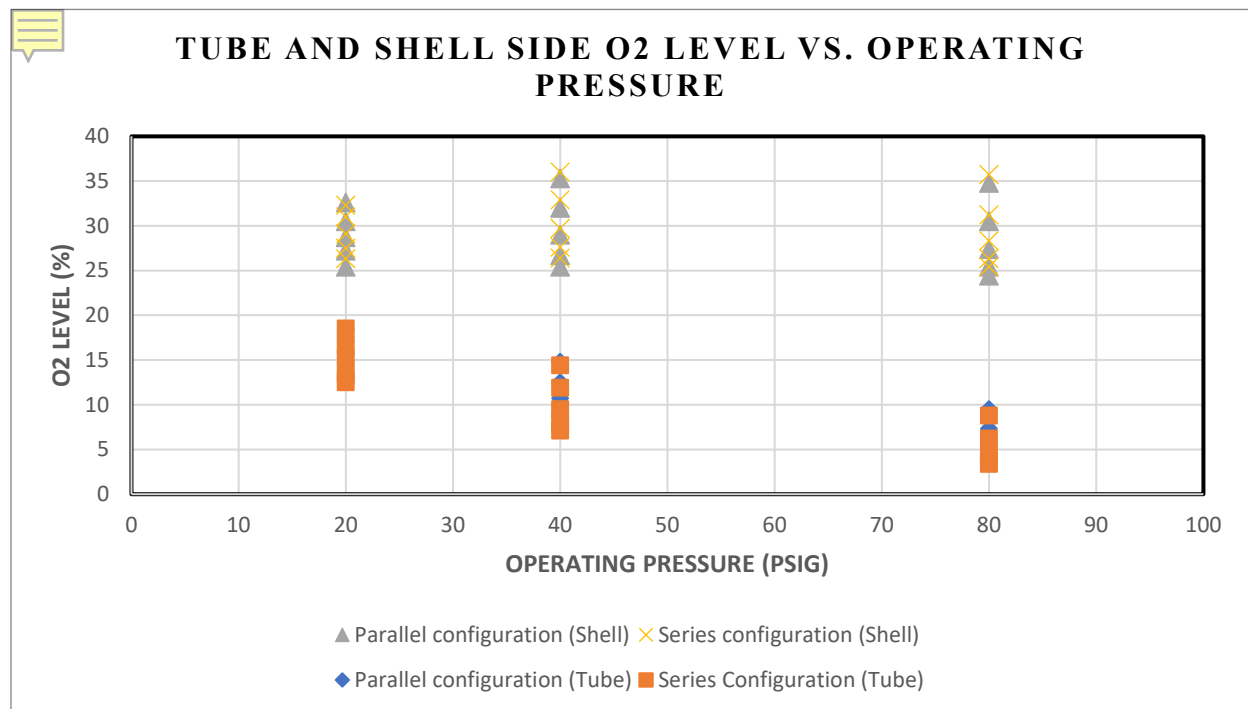


Figure 1. Oxygen levels through shell and tube side as a function of operating pressure

Figure 2. illustrates the transport coefficient of oxygen and nitrogen as a function of operating pressure. It can be observed that the transport coefficient of O_2 much higher as compared to the transport coefficient of N_2 . This was expected since the oxygen's rate of permeation was found to be higher when plotting Figure 1 which indicates that overall mass transfer coefficient of oxygen is higher than nitrogen. This observation was further confirmed through the calculations made for the ratio of transport coefficients. The values were found to be in the range from 2.60 to 4.10. The said observations can be shown in table A1.7 and Table A1.8.

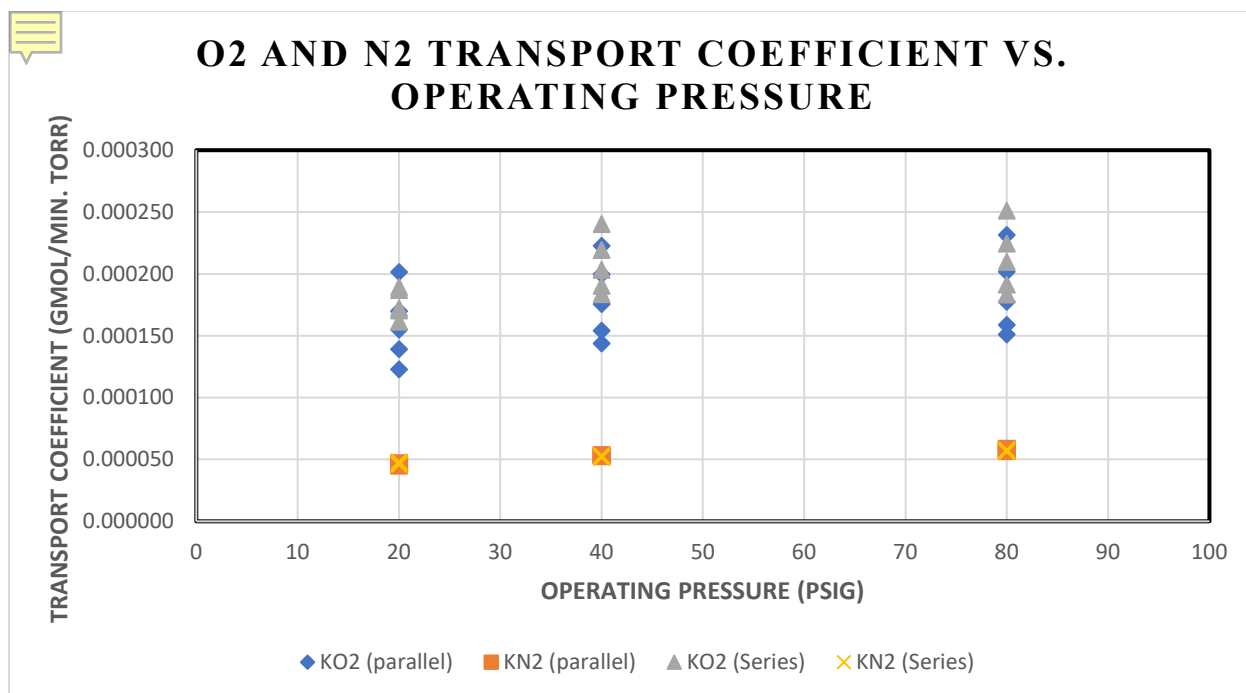


Figure 2. Transport coefficient of oxygen and nitrogen as a function of operating pressure

Figure 3. clearly illustrates that as the operating pressure increases, the ratio of transport coefficient also increases. Also, this relation was found to be inversely related with the flowrate where the transport coefficient ratio was observed to decrease to some extent when the operating pressure was increased. Selectivity was also tabulated, and it is observed that the calculations made are consistent with the results of Figure 3. i.e. when the flow of tube side increases, the selectivity also increases, and decreases with the increase in operating pressure. The results of the calculations are shown in Table A1.12 and Table A1.13.

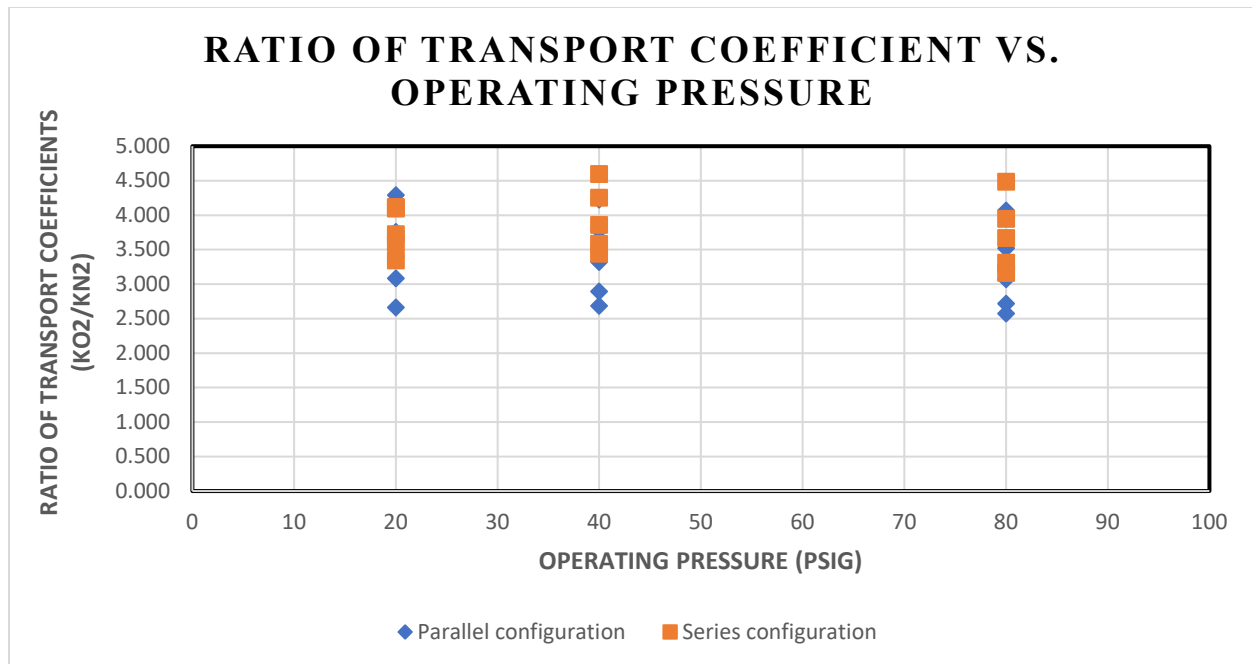


Figure 3. Ratio of transport coefficient as a function of operating pressure (Tube side flow)

Separation factor was also evaluated as a function of operating pressure. It was calculated based on the composition of the feed and retentate. **Figure 4.** Illustrates that as the operating pressure and flow rate increased, the separation factor also increased. It was observed that for the series configuration, separation factor (retentate) was higher as compared to the parallel configuration. On the other hand, it was found that the separation factor (feed) was not a significant function of operating pressure. However, feed composition separation factor was observed to be increased with the flow rate (tube side). Therefore, it was concluded from these observations that in series configuration, the separation was higher.

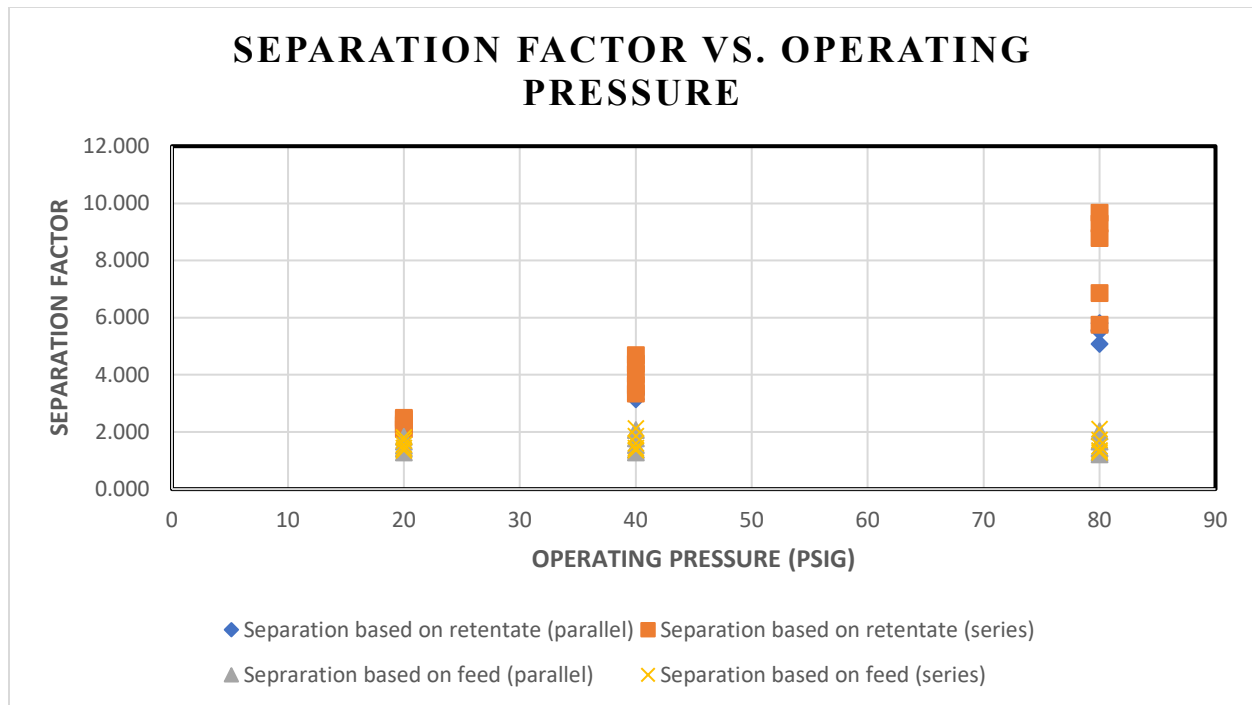


Figure 4. Separation factor as a function of operating pressure (Tube side flow)

Figure 5. Illustrates how the variation in operating function varies recoveries of nitrogen and oxygen. It was observed that for parallel and series configuration the oxygen recovery increased. Also, another observation can be made from Figure 5, that the recovery of oxygen was found to be a little higher for the configuration in series. Also, for parallel and series configuration the recovery of nitrogen was observed to decrease as the operating pressure went up. No significant difference was observed between series and parallel configuration for nitrogen recovery.

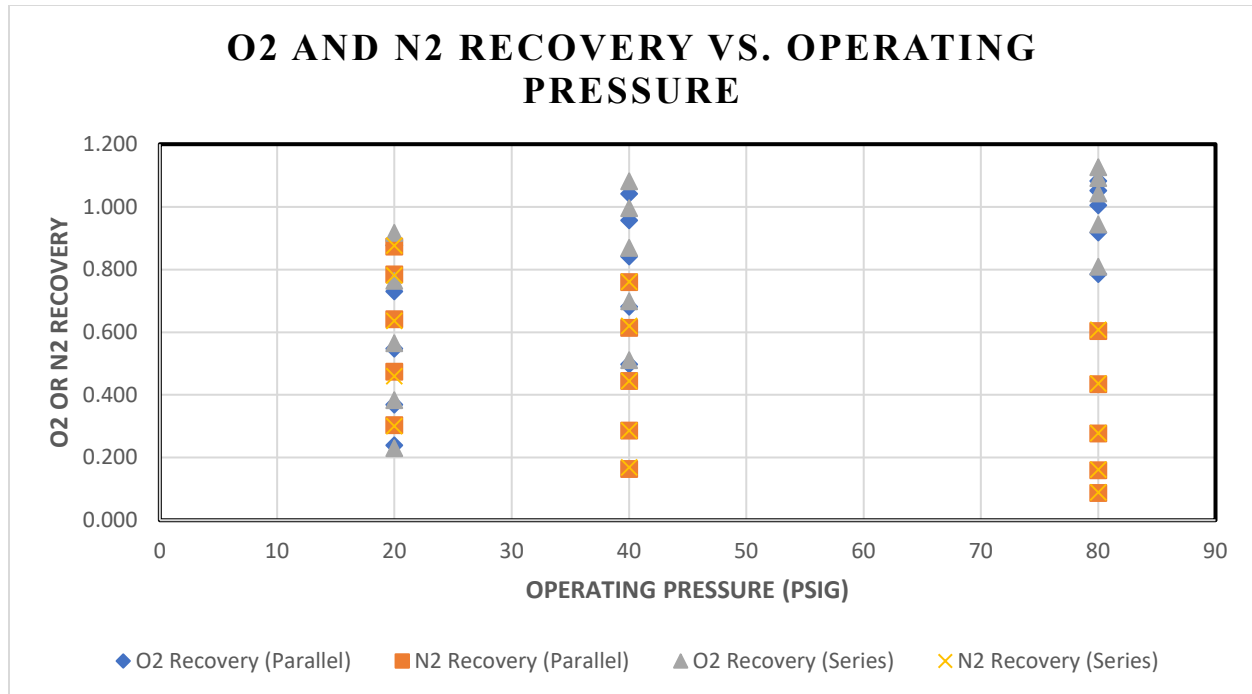


Figure 5. Recovery of oxygen and nitrogen as a function of operating pressure

It can be observed from **Figure 11** and **Figure 12**. that as the flow rate increased (tube side) for the series and parallel configuration, the recovery of oxygen decreases and recovery of nitrogen increases. This signifies the fact that ability of nitrogen to permeate decreases as operating pressure increases and the opposite is true for the oxygen permeability.

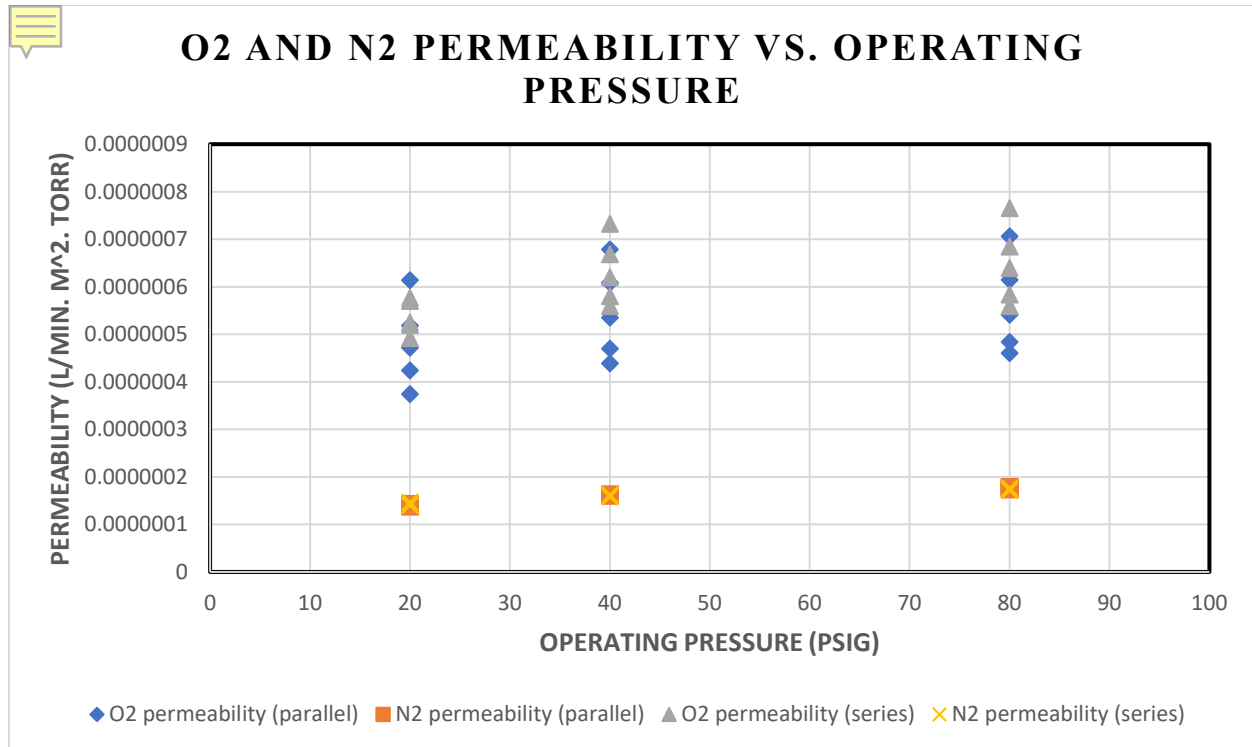


Figure 6. Permeability of oxygen and nitrogen as a function of operating pressure

From the **Table A1.12**, the results for permeability of oxygen and nitrogen in parallel configuration ranged from $3.7\text{e-}07$ - $7.1\text{E-}07$ L/min. m².torr and $1.407\text{E-}07$ - $1.735\text{E-}07$ L/min. m².torr respectively. Similarly, from **Table A1.13**, permeability of oxygen and nitrogen in series configuration ranged from $4.918\text{E-}07$ - $7.656\text{E-}07$ L/min. m².torr and $1.471\text{E-}07$ - $1.734\text{E-}07$ L/min. m².torr respectively. It can be clearly concluded that oxygen permeability is higher than that of nitrogen especially in series configuration. Effect of series and parallel configuration is not significant for the nitrogen permeability.

Appendix I: Data Tabulation/Graphs

Table A1.5 Collected data for different flow rates at different operating pressures and oxygen/nitrogen levels for series configuration

-	-	Tube Side		Shell Side		Tube Side			Shell Side		
<u>Ru</u> <u>ns</u>	<u>Press</u> <u>ure</u> <u>(psig)</u>	<u>Flow</u> <u>rate</u> <u>(L/mi</u> <u>n)</u>	<u>Oxy</u> <u>gen</u> <u>level</u> <u>(%)</u>	<u>Flow</u> <u>rate</u> <u>(L/mi</u> <u>n)</u>	<u>Oxy</u> <u>gen</u> <u>level</u> <u>(%)</u>	<u>Oxyg</u> <u>en</u> <u>Press</u> <u>ure</u> <u>(torr)</u>	<u>Nitro</u> <u>gen</u> <u>Press</u> <u>ure</u> <u>(torr)</u>	<u>Flowr</u> <u>ate</u> <u>(mol/</u> <u>min)</u>	<u>Oxyg</u> <u>en</u> <u>Press</u> <u>ure</u> <u>(torr)</u>	<u>Nitro</u> <u>gen</u> <u>Press</u> <u>ure</u> <u>(torr)</u>	<u>Flowr</u> <u>ate</u> <u>(mol/</u> <u>min)</u>
1	20	0.51	12.5	1.37	26.3	299.6 0	1494. 39	0.023	199.8 8	560.1 2	0.061
2	20	0.99	13.4	1.37	27.5	307.6 7	1486. 32	0.044	209.0 0	551.0 0	0.061
3	20	1.99	15.2	1.36	29.1	323.8 2	1470. 17	0.089	221.1 6	538.8 4	0.061
4	20	3.99	16.7	1.39	31	337.2 7	1456. 72	0.178	235.6 0	524.4 0	0.062
5	20	7.99	18.5	1.4	32.3	353.4 2	1440. 57	0.357	245.4 8	514.5 2	0.063
1	40	0.51	7.1	3.04	26.4	395.9 2	2432. 07	0.023	200.6 4	559.3 6	0.136
2	40	1	8	3.07	27.6	408.6 4	2419. 35	0.045	209.7 6	550.2 4	0.137
3	40	1.99	9.5	3.13	29.7	429.8 5	2398. 14	0.089	225.7 2	534.2 8	0.140
4	40	4	11.9	3.19	32.9	463.7 9	2364. 20	0.179	250.0 4	509.9 6	0.142
5	40	7.99	14.4	3.37	36	499.1 4	2328. 85	0.357	273.6 0	486.4 0	0.150
1	80	0.51	3.4	6.5	25.4	594.8 6	4301. 13	0.023	193.0 4	566.9 6	0.290
2	80	1	3.7	6.56	26.3	602.2 1	4293. 78	0.045	199.8 8	560.1 2	0.293
3	80	1.99	4.3	6.67	28.3	616.8 9	4279. 10	0.089	215.0 8	544.9 2	0.298
4	80	4	6.2	6.87	31.2	663.4 1	4232. 58	0.179	237.1 2	522.8 8	0.307
5	80	7.99	8.8	7.18	35.7	727.0 5	4168. 94	0.357	271.3 2	488.6 8	0.321

Table A1.6 Collected data for different flow rates at different operating pressures and oxygen/nitrogen levels for parallel configuration

Run	Press re (sig)	Tube Side		Shell Side		Tube Side			Shell Side		
		Flow ate l/min)	Oxy en vel (%)	Flow ate l/min)	Oxy en vel (%)	Oxyg n ressur (torr)	Nitroge n ressure (torr)	Flo rate mol/m (l)	Oxyg n ressur (torr)	Nitro en ressur (torr)	Flo rate mol/m (l)
1	20	0.49	13.7	1.27	25.4	310.3	1483.6	0.02	193.0	566.9	0.05
2	20	1	14.6	1.28	27.2	318.4	1475.5	0.04	206.7	553.2	0.05
3	20	1.99	15.6	1.32	28.7	327.4	1466.5	0.08	218.1	541.8	0.05
4	20	3.99	17	1.35	30.5	339.9	1454.0	0.17	231.8	528.2	0.06
5	20	8	18.4	1.45	32.6	352.5	1441.4	0.35	247.7	512.2	0.06
1	40	0.49	9.2	2.95	25.4	425.6	2402.3	0.02	193.0	566.9	0.13
2	40	1	9.8	2.99	26.7	434.1	2393.8	0.04	202.9	557.0	0.13
3	40	1.99	10.7	3.07	29	446.8	2381.1	0.08	220.4	539.6	0.13
4	40	3.99	12.5	3.2	32	472.2	2355.7	0.17	243.2	516.8	0.14
5	40	7.99	14.8	3.34	35.3	504.8	2323.1	0.35	268.2	491.7	0.14
1	80	0.5	5.5	6.39	24.4	646.2	4249.7	0.02	185.4	574.5	0.28
2	80	1	5.8	6.45	25.4	653.6	4242.3	0.04	193.0	566.9	0.28
3	80	2	6.1	6.57	27.4	660.9	4235.0	0.08	208.2	551.7	0.29
4	80	4	7.3	6.79	30.5	690.3	4205.6	0.17	231.8	528.2	0.30
5	80	8	9.5	7.15	34.8	744.1	4151.8	0.35	264.4	495.5	0.31

Table A1.7 Calculations of Permeation Rate and Transport Coefficient at Parallel

Runs	Oxygen Feed rate (mol/min)	Nitrogen Feed rate (mol/min)	Oxygen Permeation rate (mol/min)	Nitrogen Permeation rate (mol/min)	Oxygen Transport coefficient (mol O ₂ /min torr)	Nitrogen Transport coefficient (gmol N ₂ /min torr)	Ratio of Mass Transport Coefficients
1	0.016	0.062	0.014	0.042	0.000123	4.614E-05	2.660
2	0.021	0.081	0.016	0.042	0.000139	4.511E-05	3.085
3	0.031	0.117	0.017	0.042	0.000155	4.544E-05	3.406
4	0.050	0.189	0.018	0.042	0.000170	4.524E-05	3.756
5	0.088	0.334	0.021	0.044	0.000201	4.695E-05	4.290
1	0.032	0.121	0.033	0.098	0.000144	5.353E-05	2.687
2	0.037	0.141	0.036	0.098	0.000154	5.327E-05	2.894
3	0.047	0.179	0.040	0.097	0.000176	5.284E-05	3.322
4	0.067	0.254	0.046	0.097	0.000200	5.283E-05	3.778
5	0.106	0.400	0.053	0.096	0.000223	5.267E-05	4.225
1	0.064	0.243	0.070	0.216	0.000151	5.868E-05	2.574
2	0.070	0.263	0.073	0.215	0.000159	5.844E-05	2.717
3	0.080	0.303	0.080	0.213	0.000178	5.781E-05	3.071
4	0.101	0.381	0.092	0.211	0.000202	5.729E-05	3.520
5	0.141	0.535	0.111	0.208	0.000232	5.692E-05	4.068

Table A1.8 Calculations of Permeation Rate and Transport Coefficient at Series

Run s	Oxygen Feed rate (mol/min)	Nitrogen Feed rate (mol/min)	Oxygen Permeation rate (mol/min)	Nitrogen Permeation rate (mol/min)	Oxygen Transport coefficient (mol O₂/min torr)	Nitrogen Transport coefficient (gmol N₂/min torr)	Ratio of Mass Transport Coefficients
1	0.018	0.066	0.016	0.045	0.000161	4.825E-05	3.343
2	0.022	0.083	0.017	0.044	0.000170	4.741E-05	3.596
3	0.031	0.118	0.018	0.043	0.000172	4.622E-05	3.724
4	0.050	0.190	0.019	0.043	0.000189	4.593E-05	4.120
5	0.088	0.332	0.020	0.042	0.000187	4.569E-05	4.093
1	0.033	0.125	0.036	0.100	0.000183	5.334E-05	3.440
2	0.038	0.144	0.038	0.099	0.000190	5.309E-05	3.583
3	0.048	0.181	0.042	0.098	0.000203	5.270E-05	3.857
4	0.067	0.254	0.047	0.096	0.000219	5.153E-05	4.253
5	0.106	0.401	0.054	0.096	0.000240	5.226E-05	4.595
1	0.065	0.248	0.074	0.216	0.000183	5.797E-05	3.164
2	0.071	0.267	0.077	0.216	0.000191	5.781E-05	3.312
3	0.081	0.306	0.084	0.213	0.000210	5.717E-05	3.668
4	0.101	0.384	0.096	0.211	0.000224	5.688E-05	3.946
5	0.142	0.536	0.114	0.206	0.000251	5.600E-05	4.484

Table A1.9 Calculations of Permeation Mol Fraction and Separation Factor at Parallel

Runs	Mole fraction Oxygen in retentate	Mole fraction Oxygen in permeate	Mole fraction Nitrogen in retentate	Mole fraction Nitrogen in permeate	Separation factor [retentate composition (α_{AB})]	Separation factor [feed composition (α_{AB})]
1	0.137	0.254	0.863	0.746	2.145	1.289
2	0.146	0.272	0.854	0.728	2.185	1.414
3	0.156	0.287	0.844	0.713	2.178	1.523
4	0.17	0.305	0.83	0.695	2.143	1.661
5	0.184	0.326	0.816	0.674	2.145	1.831
1	0.092	0.254	0.908	0.746	3.360	1.289
2	0.098	0.267	0.902	0.733	3.353	1.379
3	0.107	0.29	0.893	0.71	3.409	1.546
4	0.125	0.32	0.875	0.68	3.294	1.781
5	0.148	0.353	0.852	0.647	3.141	2.065
1	0.055	0.244	0.945	0.756	5.545	1.222
2	0.058	0.254	0.942	0.746	5.530	1.289
3	0.061	0.274	0.939	0.726	5.810	1.428
4	0.073	0.305	0.927	0.695	5.573	1.661
5	0.095	0.348	0.905	0.652	5.085	2.020

Table A1.10 Calculations of Permeation Mol Fraction and Separation Factor at Series

Runs	Mole fraction Oxygen in retentate	Mole fraction Oxygen in permeate	Mole fraction Nitrogen in retentate	Mole fraction Nitrogen in permeate	Separation factor [retentate composition (α_{AB})]	Separation factor [feed composition (α_{AB})]
1	0.125	0.263	0.875	0.737	2.498	1.351
2	0.134	0.275	0.866	0.725	2.451	1.436
3	0.152	0.291	0.848	0.709	2.290	1.553
4	0.167	0.31	0.833	0.69	2.241	1.700
5	0.185	0.323	0.815	0.677	2.102	1.806
1	0.071	0.264	0.929	0.736	4.693	1.358
2	0.08	0.276	0.92	0.724	4.384	1.443
3	0.095	0.297	0.905	0.703	4.025	1.599
4	0.119	0.329	0.881	0.671	3.630	1.856
5	0.144	0.36	0.856	0.64	3.344	2.129
1	0.034	0.254	0.966	0.746	9.674	1.289
2	0.037	0.263	0.963	0.737	9.288	1.351
3	0.043	0.283	0.957	0.717	8.784	1.494
4	0.062	0.312	0.938	0.688	6.861	1.716
5	0.088	0.357	0.912	0.643	5.754	2.101

Table A1.10 Calculations of Permeation Flow Rate, Stage Cut, Retentate Mole Fraction & Recovery at Parallel

Runs	Retentate flowrate (std. L/min)	Permeate flowrate (std. L/min)	Stage Cut (Θ)	Feed flowrate (L/min)	Mole fraction Oxygen in permeate	Mole fraction Nitrogen in retentate	Oxygen Recovery	Nitrogen Recovery
1	0.49	1.27	0.722	1.760	0.254	0.863	0.877	0.304
2	1	1.28	0.561	2.280	0.272	0.854	0.731	0.474
3	1.99	1.32	0.399	3.310	0.287	0.844	0.548	0.641
4	3.99	1.35	0.253	5.340	0.305	0.830	0.369	0.784
5	8	1.45	0.153	9.450	0.326	0.816	0.239	0.873
1	0.49	2.95	0.858	3.440	0.254	0.908	1.042	0.164
2	1	2.99	0.749	3.990	0.267	0.902	0.957	0.286
3	1.99	3.07	0.607	5.060	0.290	0.893	0.842	0.444
4	3.99	3.2	0.445	7.190	0.320	0.875	0.681	0.614
5	8	3.34	0.295	11.340	0.353	0.852	0.497	0.760
1	0.5	6.39	0.927	6.890	0.244	0.945	1.083	0.087
2	1	6.45	0.866	7.450	0.254	0.942	1.052	0.160
3	2	6.57	0.767	8.570	0.274	0.939	1.005	0.277
4	4	6.79	0.629	10.790	0.305	0.927	0.918	0.434
5	8	7.15	0.472	15.150	0.348	0.905	0.786	0.604

Table A1.11 Calculations of Permeation Flow Rate, Stage Cut, Retentate Mole Fraction & Recovery at Series

Runs	Retentate flowrate (std. L/min)	Permeate flowrate (std. L/min)	Stage Cut (Θ)	Feed flowrate (L/min)	Mole fraction Oxygen in permeate	Mole fraction Nitrogen in retentate	Oxygen Recovery	Nitrogen Recovery
1	0.51	1.37	0.729	1.880	0.263	0.875	0.917	0.300
2	0.99	1.37	0.581	2.360	0.275	0.866	0.764	0.459
3	1.99	1.36	0.406	3.350	0.291	0.848	0.565	0.637
4	3.99	1.39	0.258	5.380	0.310	0.833	0.383	0.781
5	7.99	1.4	0.149	9.390	0.323	0.815	0.230	0.877
1	0.51	3.04	0.856	3.550	0.264	0.929	1.082	0.169
2	1	3.07	0.754	4.070	0.276	0.920	0.996	0.286
3	1.99	3.13	0.611	5.120	0.297	0.905	0.869	0.445
4	4	3.19	0.444	7.190	0.329	0.881	0.698	0.620
5	7.99	3.37	0.297	11.360	0.360	0.856	0.511	0.761
1	0.51	6.5	0.927	7.010	0.254	0.966	1.127	0.089
2	1	6.56	0.868	7.560	0.263	0.963	1.092	0.161
3	1.99	6.67	0.770	8.660	0.283	0.957	1.043	0.278
4	4	6.87	0.632	10.870	0.312	0.938	0.943	0.436
5	7.99	7.18	0.473	15.170	0.357	0.912	0.808	0.607

Table A1.12 Calculations of Flow Rate, Flux, Permeability & Selectivity at Parallel

Run	Flowrate of Oxygen in permeate (m ³ /s)	Flowrate of Nitrogen in permeate (m ³ /s)	Flux of Oxygen (mol/m ² s)	Flux of Nitrogen (mol/m ²)	Oxygen Permeability (L/min. m ² .torr)	Nitrogen Permeability (L/min. m ² .torr)	Selectivity (α)
1	5.376E-06	1.579E-05	4.391E-05	0.000129	3.742E-07	1.407E-07	2.660
2	5.803E-06	1.553E-05	4.739E-05	0.000127	4.242E-07	1.375E-07	3.085
3	6.314E-06	1.569E-05	5.156E-05	0.000128	4.718E-07	1.385E-07	3.406
4	6.863E-06	1.564E-05	5.604E-05	0.000128	5.182E-07	1.379E-07	3.756
5	7.878E-06	1.629E-05	6.434E-05	0.000133	6.142E-07	1.432E-07	4.290
1	1.249E-05	3.668E-05	1.020E-04	0.000300	4.385E-07	1.632E-07	2.687
2	1.331E-05	3.653E-05	1.087E-04	0.000298	4.700E-07	1.624E-07	2.894
3	1.484E-05	3.633E-05	1.212E-04	0.000297	5.352E-07	1.611E-07	3.322
4	1.707E-05	3.627E-05	1.394E-04	0.000296	6.084E-07	1.611E-07	3.778
5	1.965E-05	3.602E-05	1.605E-04	0.000294	6.785E-07	1.606E-07	4.225
1	2.599E-05	8.051E-05	2.122E-04	0.000658	4.605E-07	1.789E-07	2.574
2	2.731E-05	8.020E-05	2.230E-04	0.000655	4.842E-07	1.782E-07	2.717
3	3.000E-05	7.950E-05	2.450E-04	0.000649	5.412E-07	1.763E-07	3.071
4	3.452E-05	7.865E-05	2.819E-04	0.000642	6.147E-07	1.747E-07	3.520
5	4.147E-05	7.770E-05	3.387E-04	0.000635	7.060E-07	1.735E-07	4.068

Table A1.13 Calculations of Flow Rate, Flux, Permeability & Selectivity at Series

Run	Flowrate of Oxygen in permeate (m ³ /s)	Flowrate of Nitrogen in permeate (m ³ /s)	Flux of Oxygen (mol/m ² s)	Flux of Nitrogen (mol/m ²)	Oxygen Permeability (L/min. m ² .torr)	Nitrogen Permeability (L/min. m ² .torr)	Selectivity (α)
1	6.005E-06	1.683E-05	4.904E-05	0.000137	4.918E-07	1.471E-07	3.343
2	6.279E-06	1.655E-05	5.128E-05	0.000135	5.197E-07	1.445E-07	3.596
3	6.596E-06	1.607E-05	5.387E-05	0.000131	5.247E-07	1.409E-07	3.724
4	7.182E-06	1.599E-05	5.865E-05	0.000131	5.769E-07	1.400E-07	4.120
5	7.537E-06	1.580E-05	6.155E-05	0.000129	5.702E-07	1.393E-07	4.093
6	1.338E-05	3.729E-05	1.092E-04	0.000305	5.594E-07	1.626E-07	3.440
7	1.412E-05	3.704E-05	1.153E-04	0.000303	5.799E-07	1.619E-07	3.583
8	1.549E-05	3.667E-05	1.265E-04	0.000299	6.198E-07	1.607E-07	3.857
9	1.749E-05	3.567E-05	1.428E-04	0.000291	6.683E-07	1.571E-07	4.253
10	2.022E-05	3.595E-05	1.651E-04	0.000294	7.322E-07	1.593E-07	4.595
11	2.752E-05	8.082E-05	2.247E-04	0.000660	5.593E-07	1.767E-07	3.164
12	2.875E-05	8.058E-05	2.348E-04	0.000658	5.837E-07	1.763E-07	3.312
13	3.146E-05	7.971E-05	2.569E-04	0.000651	6.394E-07	1.743E-07	3.668
14	3.572E-05	7.878E-05	2.917E-04	0.000643	6.844E-07	1.734E-07	3.946
15	4.272E-05	7.695E-05	3.489E-04	0.000628	7.656E-07	1.707E-07	4.484

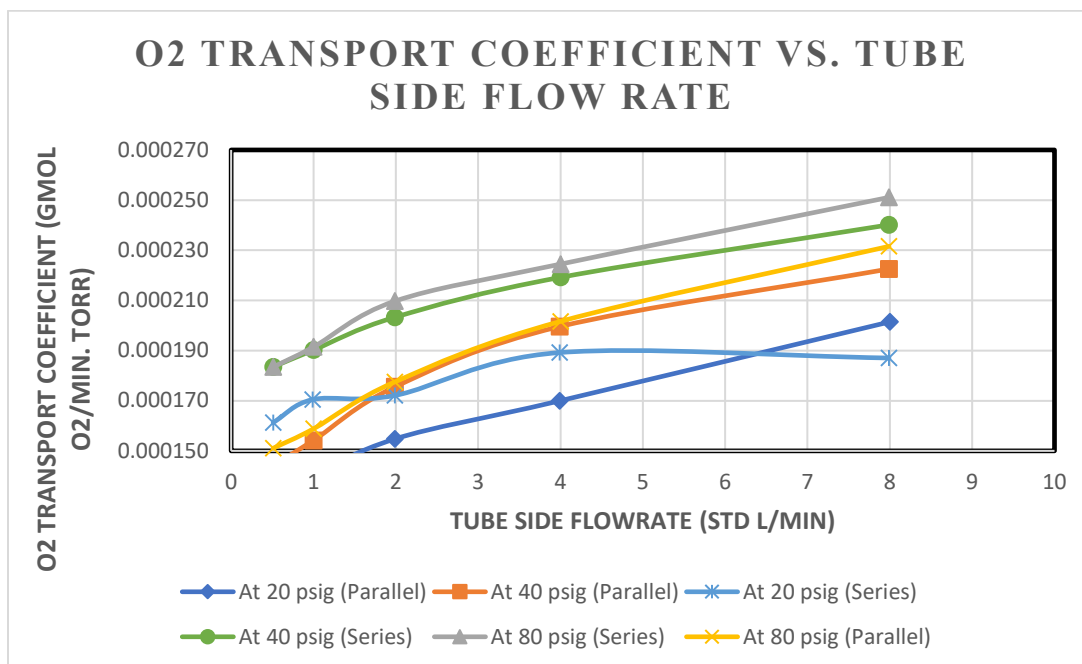


Figure 7. Transport coefficient of oxygen (tube side flow)

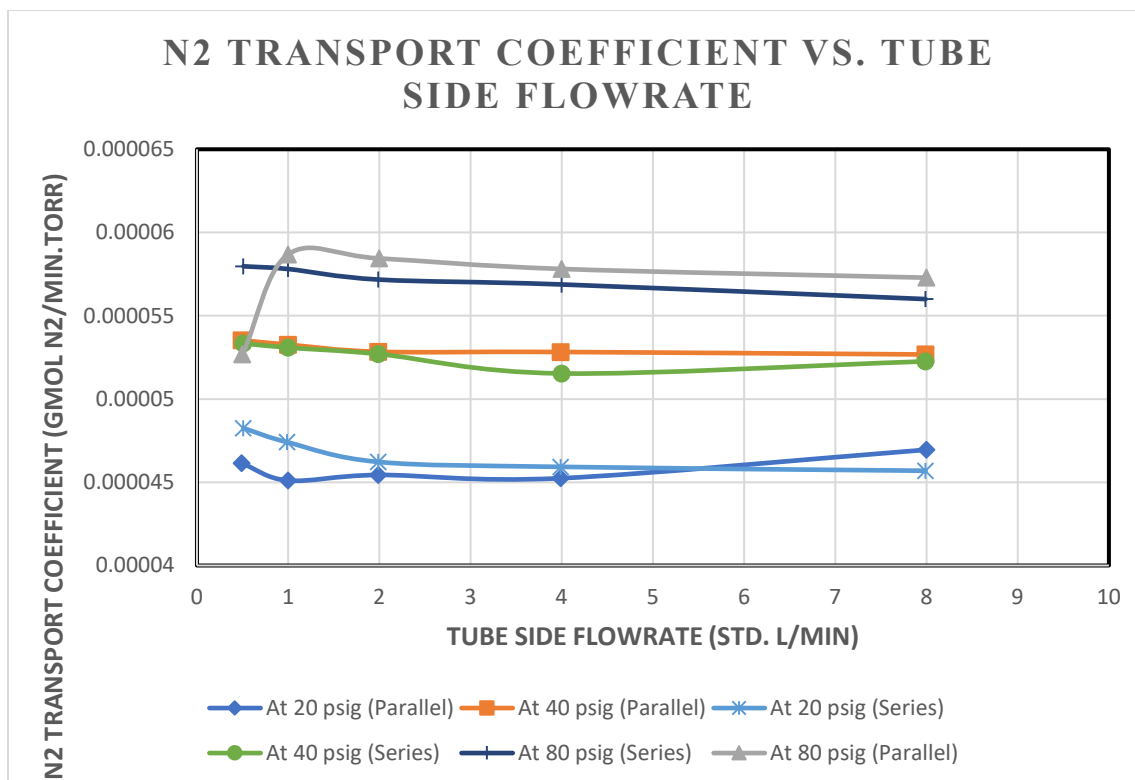


Figure 8. Transport coefficient of nitrogen (tube side flow)

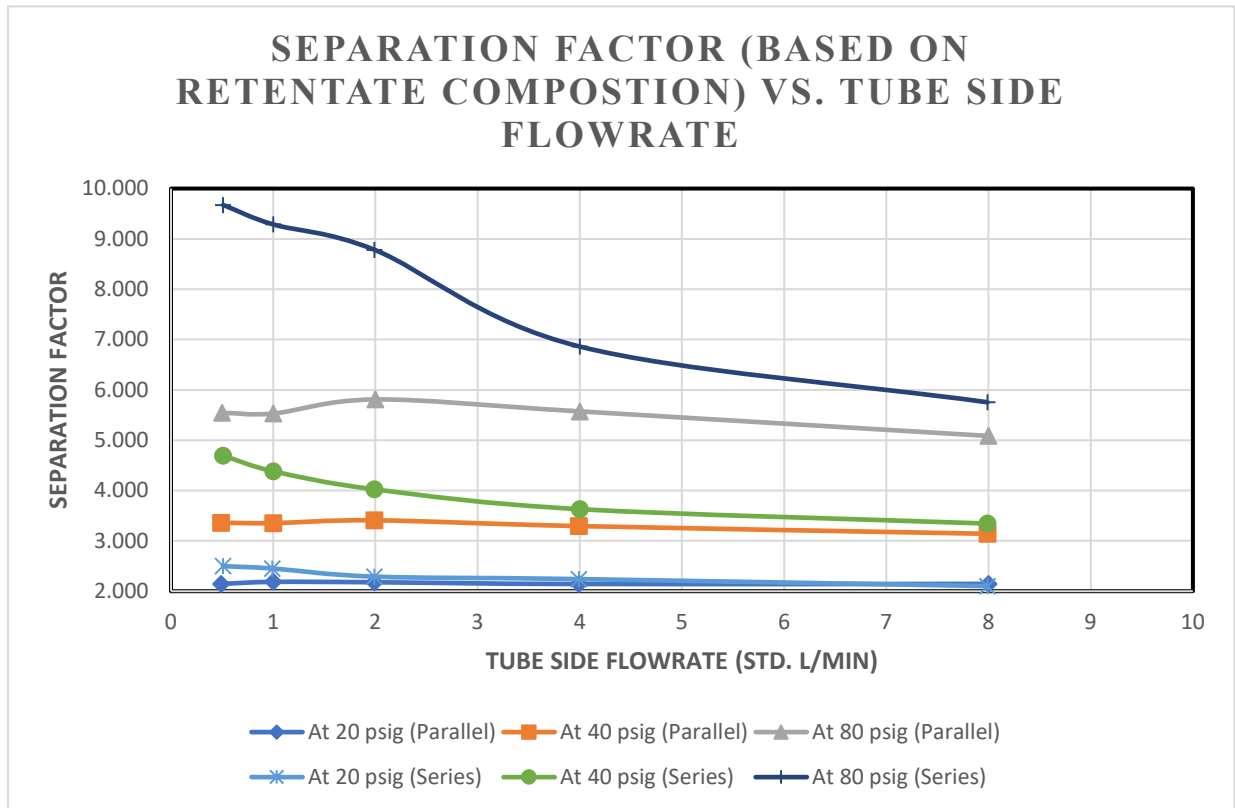


Figure 9. Separation factor based of retentate (tube side flow)

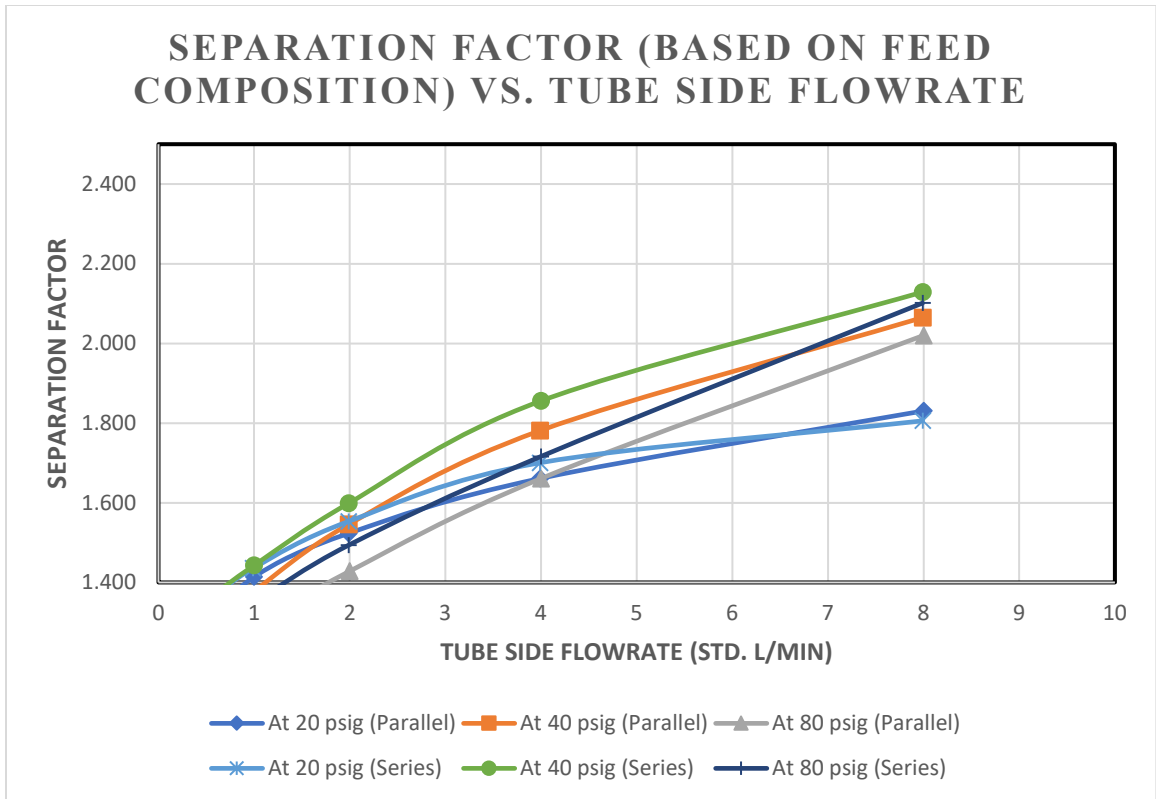


Figure 10. Separation factor based of feed (tube side flow)

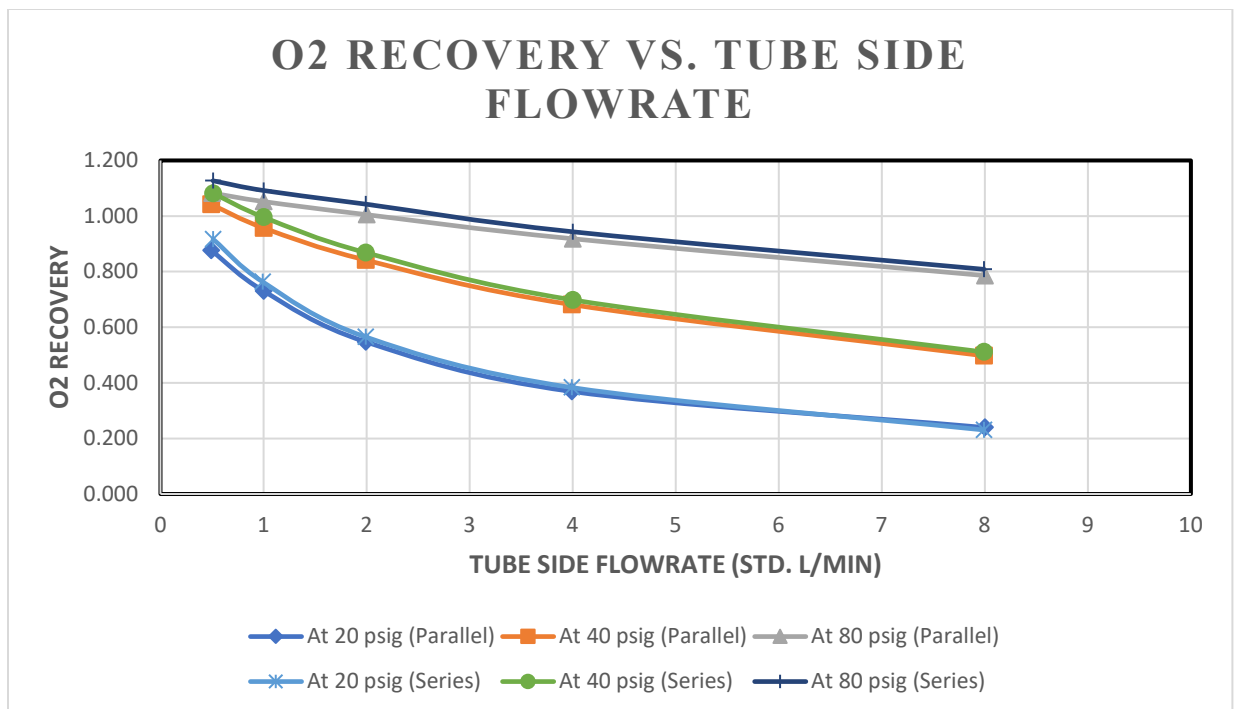


Figure 11. Recovery of oxygen (tube side flow)

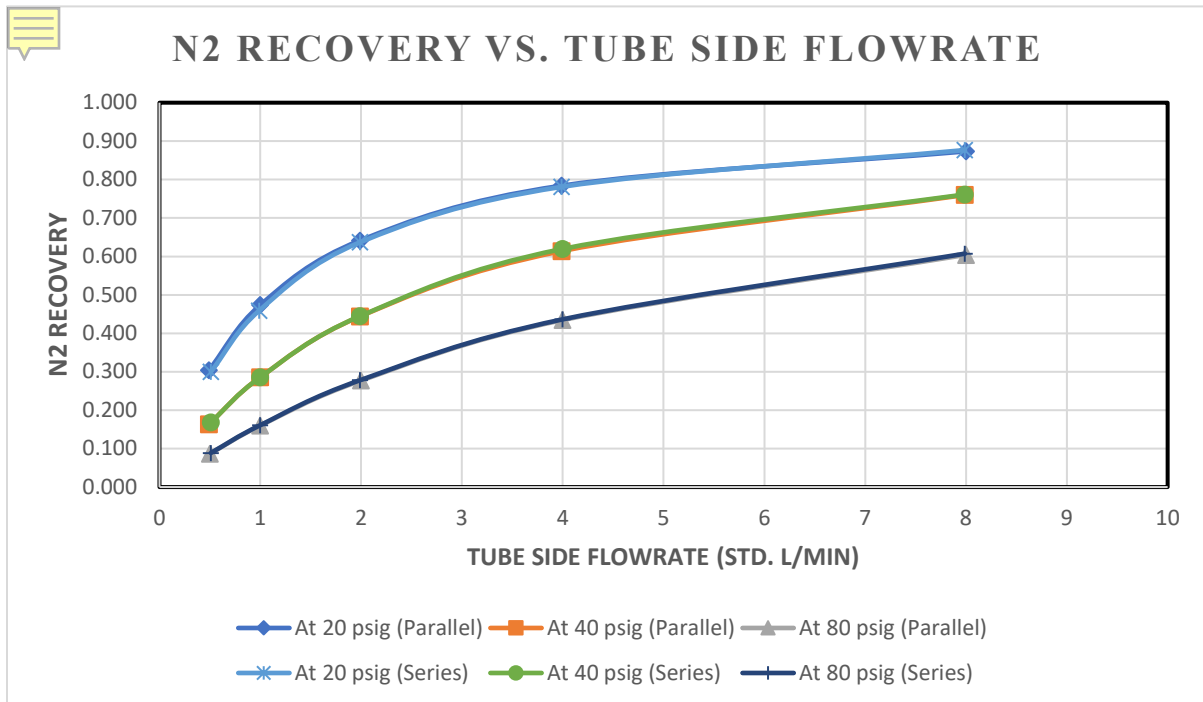


Figure 12. Recovery of nitrogen (tube side flow)

Appendix II: Error Analysis:

Mean Deviation

Table 1.1 Deviations at 20 PSIG Parallel

Readings	20 Psig			
	Flowrate (SPLM) Tube side	O2 %	Flow rate (SLPM) shell surface	O2 %
1	2.604	2.16	0.064	3.48
2	2.094	1.26	0.054	1.68
3	1.104	0.26	0.014	0.18
4	0.896	1.14	0.016	1.62
5	4.906	2.54	0.116	3.72

Table 1.2 Deviations at 40 PSIG Parallel

Readings	40 Psig			
	Flowrate (SPLM) Tube side	O2 %	Flow rate (SLPM) shell surface	O2 %
1	2.602	2.2	0.16	4.28
2	2.092	1.6	0.12	2.98
3	1.102	0.7	0.04	0.68
4	0.898	1.1	0.09	2.32
5	4.898	3.4	0.23	5.62

Table 1.3 Deviations at 80 PSIG Parallel

Readings	80 Psig			
	Flowrate (SPLM) Tube side	O2 %	Flow rate (SLPM) shell surface	O2 %
1	2.6	1.34	0.28	4.1
2	2.1	1.04	0.22	3.1
3	1.1	0.74	0.1	1.1
4	0.9	0.46	0.12	2
5	4.9	2.66	0.48	6.3

Table 1.4 Deviations at 20 PSIG Series

Readings	20 Psig			
	Flowrate (SPLM) Tube side	O2 %	Flow rate (SLPM) shell surface	O2 %
1	2.584	2.76	0.008	2.94
2	2.104	1.86	0.008	1.74
3	1.104	0.06	0.018	0.14
4	0.896	1.44	0.012	1.76
5	4.896	3.24	0.022	3.06

Table 1.5 Deviations at 40 PSIG Series

Readings	40 Psig			
	Flowrate (SPLM) Tube side	O2 %	Flow rate (SLPM) shell surface	O2 %
1	2.588	3.08	0.12	4.12
2	2.098	2.18	0.09	2.92
3	1.108	0.68	0.03	0.82
4	0.902	1.72	0.03	2.38
5	4.892	4.22	0.21	5.48

Table 1.6 Deviations at 80 PSIG Series

Readings	80 Psig			
	Flowrate (SPLM) Tube side	O2 %	Flow rate (SLPM) shell surface	O2 %
1	2.588	1.88	0.256	3.98
2	2.098	1.58	0.196	3.08
3	1.108	0.98	0.086	1.08
4	0.902	0.92	0.114	1.82
5	4.892	3.52	0.424	6.32

Averages

Table 2.1 Averages at 20 PSIG Paral

	20 Psig			
	Flowrate (SPLM) Tube side	O2 %	Flow rate (SLPM) shell surface	O2 %
average	3.094	15.86	1.334	28.88

Table 2.2 Averages at 40 PSIG Paral

	40 Psig			
	Flowrate (SPLM) Tube side	O2 %	Flow rate (SLPM) shell surface	O2 %
average	3.092	11.4	3.11	29.68

Table 2.3 Averages at 80 PSIG Paral

	80 Psig			
	Flowrate (SPLM) Tube side	O2 %	Flow rate (SLPM) shell surface	O2 %
average	3.1	6.84	6.67	28.5

Table 2.4 Averages at 20 PSIG
Series

	20 Psig			
	Flowrate (SPLM) Tube side	O2 %	Flow rate (SLPM) shell surface	O2 %
average	3.094	15.26	1.378	29.24

Table 2.5 Averages at 40 PSIG
Series

	40 Psig			
	Flowrate (SPLM) Tube side	O2 %	Flow rate (SLPM) shell surface	O2 %
average	3.098	10.18	3.16	30.52

Table 2.6 Averages at 80 PSIG
Series

	80 Psig			
	Flowrate (SPLM) Tube side	O2 %	Flow rate (SLPM) shell surface	O2 %
average	3.098	5.28	6.756	29.38

Ranges

Table 3.1 Range at 20 PSIG Parallel

	20 Psig			
	Flowrate (SPLM) Tube side	O2 %	Flow rate (SLPM) shell surface	O2 %
Min	0.49	13.7	1.27	25.4
Max	8	18.4	1.45	32.6

Table 3.2 Range at 40 PSIG Parallel

	40 Psig			
	Flowrate (SPLM) Tube side	O2 %	Flow rate (SLPM) shell surface	O2 %
Min	0.49	9.2	2.95	25.4
Max	7.99	14.8	3.34	35.3

Table 3.3 Range at 80 PSIG Parallel

	80 Psig			
	Flowrate (SPLM) Tube side	O2 %	Flow rate (SLPM) shell surface	O2 %
Min	0.5	5.5	6.39	24.4
Max	8	9.5	7.15	34.8

Table 3.4 Range at 20 PSIG Series

	20 Psig			
	Flowrate (SPLM) Tube side	O2 %	Flow rate (SLPM) shell surface	O2 %
Min	0.51	12.5	1.36	26.3
Max	7.99	18.5	1.4	32.3

Table 3.5 Range at 40 PSIG Series

	40 Psig			
	Flowrate (SPLM) Tube side	O2 %	Flow rate (SLPM) shell surface	O2 %
Min	0.51	7.1	3.04	26.4
Max	7.99	14.4	3.37	36

Table 3.6 Range at 80 PSIG Series

	80 Psig			
	Flowrate (SPLM) Tube side	O2 %	Flow rate (SLPM) shell surface	O2 %
Min	0.51	3.4	6.5	25.4
Max	7.99	8.8	7.18	35.7

Standard Deviations

Table 4.1 Standard Deviation at 80 PSIG Parallel

	80 Psig			
	O2 mole frac in tube	N2 mole frac in tube	O2 mole frac in shell	N2 mole frac in shell
	"x1a"	"x2a"	"x1s"	"x2s"
STDEV	0.016	0.016	0.042	0.042
	O2 Permeation rate (gmol/min)	N2 Permeation rate (gmol/min)	O2 Transport Coefficient (gmol O2/(min torr)	N2 Transport Coefficient (gmol N2/(min torr)
STDEV	0.017	0.003	0.000	0.000
	Separation Coefficients	Experimental Separation Factors	Experimental Separation Factors	
	" α_{ab} "	" α'_{ab} "	" α''_{ab} "	
STDEV	0.086	1.167	0.263	

Table 4.2 Standard Deviation at 40 PSIG Parallel

	40 Psig			
	O2 mole frac in tube	N2 mole frac in tube	O2 mole frac in shell	N2 mole frac in shell
	"x1a"	"x2a"	"x1s"	"x2s"
STDEV	0.023	0.023	0.040	0.040
	O2 Permeation rate	N2 Permeation rate	O2 Transport Coefficient "K"	N2 Transport Coefficient "K"
	(gmol/min)	(gmol/min)	(gmol O2/ (min torr)	(gmol N2/ (min torr)
STDEV	0.008	0.001	0.000	0.000
	Separation Coefficients	Experimental Separation Factors	Experimental Separation Factors	
	" α_{ab} "	" α'_{ab} "	" α''_{ab} "	
STDEV	0.083	0.647	0.104	

Table 4.3 Standard Deviation at 20 PSIG Parallel

	20 Psig			
	O2 mole frac in tube	N2 mole frac in tube	O2 mole frac in shell	N2 mole frac in shell
	"x1a"	"x2a"	"x1s"	"x2s"
STDEV	0.019	0.019	0.028	0.028
	O2 Permeation rate	N2 Permeation rate	O2 Transport Coefficient "K"	N2 Transport Coefficient "K"
	(gmol/min)	(gmol/min)	(gmol O2/ (min torr)	(gmol N2/ (min torr)
STDEV	0.003	0.001	0.000	0.000
	Separation Coefficients	Experimental Separation Factors	Experimental Separation Factors	
	" α_{ab} "	" α'_{ab} "	" α''_{ab} "	
STDEV	0.056	0.296	0.021	

Table 4.4 Standard Deviation at 80 PSIG Series

	80 Psig			
	O2 mole frac in tube	N2 mole frac in tube	O2 mole frac in shell	N2 mole frac in shell
	"x1a"	"x2a"		"x2s"
STDEV	0.022	0.022	0.042	
	O2 Permeation rate	N2 Permeation rate	O2 Transport Coefficient "K"	N2 Transport Coefficient "K"
	(gmol/min)	(gmol/min)	(gmol O2/ (min torr)	(gmol N2/ (min torr)
STDEV	0.017	0.004	0.000	0.000
	Separation Coefficients	Experimental Separation Factors	Experimental Separation Factors	
	" α_{ab} "	" α'_{ab} "	" α''_{ab} "	
STDEV	0.087	0.330	1.688	

Table 4.5 Standard Deviation at 40 PSIG Series

	20 Psig			
	O2 mole frac in tube	N2 mole frac in tube	O2 mole frac in shell	N2 mole frac in shell
	"x1a"	"x2a"	"x1s"	"x2s"
STDEV	0.030	0.030	0.039	0.039
	O2 Permeation rate	N2 Permeation rate	O2 Transport Coefficient "K"	N2 Transport Coefficient "K"
	(gmol/min)	(gmol/min)	(gmol O2/ (min torr)	(gmol N2/ (min torr)
STDEV	0.007	0.002	0.000	0.000
	Separation Coefficients	Experimental Separation Factors	Experimental Separation Factors	
	" α_{ab} "	" α'_{ab} "	" α''_{ab} "	
STDEV	0.083	0.316	0.547	

Table 4.6 Standard Deviation at 20 PSIG Series

	20 Psig			
	O2 mole frac in tube	N2 mole frac in tube	O2 mole frac in shell	N2 mole frac in shell
	"x1a"	"x2a"	"x1s"	"x2s"
STDEV	0.024	0.024	0.025	0.025
	O2 Permeation rate	N2 Permeation rate	O2 Transport Coefficient "K"	N2 Transport Coefficient "K"
	(gmol/min)	(gmol/min)	(gmol O2/ (min torr)	(gmol N2/ (min torr)
STDEV	0.002	0.001	0.000	0.000
	Separation Coefficients	Experimental Separation Factors	Experimental Separation Factors	
	" α_{ab} "	" α'_{ab} "	" α''_{ab} "	
STDEV	0.049	0.186	0.161	

Table 4.7 Standard Deviation at Parallel for Permeation and Transport Coefficient

		Oxygen Permeation rate (mol/min)	Nitrogen Permeation rate (mol/min)	Oxygen Transport coefficient (mol O₂/min torr)	Nitrogen Transport coefficient (gmol N₂/min torr)	Ratio of Mass Transport Coefficients
	STDEV	0.031	0.073	0.000	0.000	0.587

Table 4.8 Standard Deviation at Series for Permeation and Transport Coefficient

	Oxygen Permeation rate (mol/min)	Nitrogen Permeation rate (mol/min)	Oxygen Transport coefficient (mol O₂/min torr)	Nitrogen Transport coefficient (gmol N₂/min torr)	Ratio of Mass Transport Coefficients
STDEV	0.032	0.073	0.000	0.000	0.432

Table 4.9 Standard Deviation at Parallel for Separation Factors & Retentate mole Fraction

	Mole fraction Nitrogen in retentate	Mole fraction Nitrogen in permeate	Separation factor [retentate composition (α_{AB})]	Separation factor [feed composition (α_{AB})]
STDEV	0.042161	0.034949	1.446082	0.269601

Table 4.10 Standard Deviation at Series for Separation Factors & Retentate mole Fraction

	Mole fraction Nitrogen in retentate	Mole fraction Nitrogen in permeate	Separation factor [retentate composition (α_{AB})]	Separation factor [feed composition (α_{AB})]
STDEV	0.048	0.034	2.675	0.268

Table 4.11 Standard Deviation at Parallel for Permeate mol Fraction and Recovery

	Stage Cut (Θ)	Mole fraction Oxygen in permeate	Mole fraction Nitrogen in retentate	Oxygen Recovery	Nitrogen Recovery
STDEV	0.239	0.035	0.042	0.261	0.246

Table 4.12 Standard Deviation at Series for Permeate mol Fraction and Recovery

	Stage Cut (Θ)	Mole fraction Oxygen in permeate	Mole fraction Nitrogen in retentate	Oxygen Recovery	Nitrogen Recovery
STDEV	0.239	0.034	0.048	0.275	0.246

Table 4.13 Standard Deviation at Parallel for Permeate Flow Rate, Flux, Permeability, and Selectivity

	Flowrate of Oxygen in permeate (m³/s)	Flowrate of Nitrogen in permeate (m³/s)	Flux of Oxygen (mol/m²s)	Flux of Nitrogen (mol/m²)	Oxygen Permeability (L/min. m².torr)	Nitrogen Permeability (L/min. m².torr)	Selectivity (α)
STDEV	1.151E-05	2.740E-05	9.401E-05	2.238E-04	9.690E-08	1.574E-08	5.866E-01

Table 4.14 Standard Deviation at Series for Permeate Flow Rate, Flux, Permeability, and Selectivity

	Flowrate of Oxygen in permeate (m³/s)	Flowrate of Nitrogen in permeate (m³/s)	Flux of Oxygen (mol/m²s)	Flux of Nitrogen (mol/m²)	Oxygen Permeability (L/min. m².torr)	Nitrogen Permeability (L/min. m².torr)	Selectivity (α)
STDEV	1.193E-05	2.725E-05	9.739E-05	2.225E-04	7.903E-08	1.375E-08	4.317E-01

Appendix III: Sample Calculations

Calculating oxygen transport coefficient:

1. Tube side operating pressure:

$$P(t) = \frac{(\text{pressure in psig} + 14.7 \text{ psia})}{14.7 \text{ psia}} \times 760 \text{ torr}$$

$$P(t) = \frac{(20 \text{ psig} + 14.7 \text{ psia})}{14.7 \text{ psia}} \times 760 \text{ torr}$$

2. Tube side flow rate:

$$F \left(\frac{\text{mol}}{\text{min}} \right) = \frac{\left[\text{Tube side Flowrate of air in } \left(\frac{\text{L}}{\text{min}} \right) \right]}{22.4 \left(\frac{\text{mol}}{\text{L}} \right)}$$

$$F \left(\frac{\text{mol}}{\text{min}} \right) = \frac{\left[0.49 \left(\frac{\text{L}}{\text{min}} \right) \right]}{22.4 \left(\frac{\text{mol}}{\text{L}} \right)} = 0.022 \frac{\text{mol}}{\text{min}}$$

3. Shell side flow rate:

$$G \left(\frac{\text{mol}}{\text{min}} \right) = \frac{\left[\text{Shell side Flowrate of air in } \left(\frac{\text{L}}{\text{min}} \right) \right]}{22.4 \left(\frac{\text{mol}}{\text{L}} \right)}$$

$$G \left(\frac{\text{mol}}{\text{min}} \right) = \frac{\left[1.27 \left(\frac{\text{L}}{\text{min}} \right) \right]}{22.4 \left(\frac{\text{mol}}{\text{L}} \right)} = 0.057 \frac{\text{mol}}{\text{min}}$$

4. Permeation rate of Oxygen:

$$P_{O_2} \left(\frac{\text{mol}}{\text{min}} \right) = \text{Shell side Flowrate of Oxygen in } \left(\frac{\text{mol}}{\text{min}} \right) \times \text{Oxygen \%level}$$

$$P_{O_2} \left(\frac{\text{mol}}{\text{min}} \right) = 0.057 \left(\frac{\text{mol}}{\text{min}} \right) \times 25.4 \% = 0.014 \frac{\text{mol}}{\text{min}}$$

5. O₂ transport coefficient:

$$K_{O_2} \left(\frac{\text{mol}}{\text{min} \cdot \text{torr}} \right) = \frac{\left[\text{Permeation rate of Oxygen } \left(\frac{\text{mol}}{\text{min}} \right) \right]}{\left[\text{Tube side } O_2 \text{ pressure} - \text{Shell side } O_2 \text{ pressure (torr)} \right]}$$

$$K_{O_2} \left(\frac{\text{mol}}{\text{min} \cdot \text{torr}} \right) = \frac{\left[0.014 \left(\frac{\text{mol}}{\text{min}} \right) \right]}{[(310.36 - 193.04) (\text{torr})]} = 0.000139 \frac{\text{mol}}{\text{min} \cdot \text{torr}}$$

6. Separation factor based on retention composition

$$\alpha'_{AB} = \frac{\frac{\chi_{Ap}}{\chi_{Bp}}}{\frac{\chi_{Ar}}{\chi_{Br}}}$$

$$\alpha'_{AB} = \frac{\frac{0.254}{0.746}}{\frac{0.137}{0.863}}$$

X_{Ap} is the mole fraction of oxygen in permeate and X_{Bp} is the mole fraction of nitrogen in permeate. X_{Ar} is the mole fraction of oxygen in retentate and X_{Br} is the mole fraction of nitrogen in retentate.

7. Separation factor based on feed composition

$$\alpha''_{AB} = \frac{\frac{\chi_{Ap}}{\chi_{Bp}}}{\frac{\chi_{Af}}{\chi_{Bf}}}$$

$$\alpha'_{AB} = \frac{\frac{0.254}{0.746}}{\frac{0.209}{0.791}}$$

X_{Af} is the mole fraction of oxygen in feed and X_{Bf} is the mole fraction of nitrogen in feed. 

Team Member Contributions

NAME: Daniela Estarita

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	3	Worked as a team
Pre-Lab Editing	0.25	Worked on intro portion
Experimental Objective (Prelab)	n/a	Split portions as a group
Apparatus (prelab)	n/a	Split portions as a group
Materials & Supplies (prelab)	n/a	Split portions as a group
Procedure (prelab)	n/a	Split portions as a group
Project Safety Evaluation (prelab)	n/a	Split portions as a group
Final Lab Editing	.25	Read all portions
Executive Summary (final lab)	n/a	Split portions as a group
Introduction (Final lab)	1	Wrote and found reference
Theory (Final Lab)	n/a	Split portions as a group
Results (final lab)	n/a	Split portions as a group
Discussion (final lab)	n/a	Split portions as a group
References (final and/or prelab)	0.15	Found 2 references for intro
Data Tabulation/Graphs (final)	5	Calculated needed coefficients, rates, fractions, etc
Error Analysis (final lab)	3	Calculated errors of coefficients, rates, fractions, etc
Sample Calculations (final Lab)	0.25	Lab manual had wrong sample calculations had to re write formulas
Power Point Presentation	0.25	Worked on intro slide
Total	13.15	

NAME: Huda Usman

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	5.5	Worked as a team
Pre-Lab Editing	1	
Experimental Objective (Prelab)	0	Helped other team member edit
Apparatus (prelab)	0.15	Helped other team member edit
Materials & Supplies (prelab)	0.30	Wrote the supplies
Procedure (prelab)	0.15	Helped other team member edit
Project Safety Evaluation (prelab)	0.25	Worked on it
Final Lab Editing	2	Made the required changes
Executive Summary (final lab)	3	Worked on it
Introduction (Final lab)	0	Assigned to other team member
Theory (Final Lab)	0	Assigned to other team member
Results (final lab)	5	Worked on it, plotted graphs, and explained the trends
Discussion (final lab)	0	Assigned to other team member
References (final and/or prelab)	0.15	Cited references used to write executive summary and results in general
Data Tabulation/Graphs (final)	0	Assigned to other team member
Error Analysis (final lab)	0	Assigned to other team member
Sample Calculations (final Lab)	2	Organized data calculation on excel sheet and provided formulas used to other team member
Power Point Presentation	n/a	n/a
Total	19.5	

NAME: Nisha Patel

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	5.5	Worked as a team
Pre-Lab Editing	1	Worked with a team member
Experimental Objective (Prelab)	0	
Apparatus (prelab)	0.30	Reviewed for team member
Materials & Supplies (prelab)	0.15	Reviewed for team member
Procedure (prelab)	2	Wrote the procedure for lab
Project Safety Evaluation (prelab)	0.15	Reviewed for team member
Final Lab Editing	2	Final editing
Executive Summary (final lab)	0	
Introduction (Final lab)	0	
Theory (Final Lab)	5	Wrote the theory of this experiment
Results (final lab)	0	
Discussion (final lab)	0	
References (final and/or prelab)	0.15	Cited references for theory
Data Tabulation/Graphs (final)	0	
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	1	Showed sample calculations of the actual calculations
Power Point Presentation	n/a	n/a
Total	15.1	

NAME: Michael Topalis

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	3	Worked as a team
Pre-Lab Editing	3	Worked as team
Experimental Objective (Prelab)	0	Team member
Apparatus (prelab)	3	They were written out and pictures were taken
Materials & Supplies (prelab)	3	It was written out
Procedure (prelab)	0	Team member
Project Safety Evaluation (prelab)	0	Team member
Final Lab Editing	1	group
Executive Summary (final lab)	0	Team member
Introduction (Final lab)	0	Team member
Theory (Final Lab)	1	Wrote a part
Results (final lab)	0	Team member
Discussion (final lab)	4	wrote
References (final and/or prelab)	1	group
Data Tabulation/Graphs (final)	0	Team member
Error Analysis (final lab)	0	Team member
Sample Calculations (final Lab)	0	Team member
Power Point Presentation	1	Group
Total	20	Worked as a team